

A Graph Neural Network Framework for Vascular Streak Dieback Recognition

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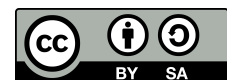
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ABSTRACT

One of the most worrying threats to cocoa production in Indonesia is Vascular Streak Dieback, or VSD for short. Early symptoms of VSD are difficult to distinguish using traditional classification techniques. One technique used to differentiate is by observing the visual appearance of cocoa leaves. One technique used to process image data is image clarification. In image clarification, convolutional neural networks, or CNNs for short, are widely used to recognize the disease. However, the CNN approach has weaknesses in the context of relying on local receptive fields, making it less able to model spatially distributed symptom indicators. This study examines a graph-based classification framework inspired by the Agri-Graph Neural Network (Agri-GNN).by converting cocoa leaf images into a structured graph representation to capture spatial relationships between leaf areas. A specialized Graph Neural Network (GNN) model was applied to perform message passing and classify leaves into healthy or VSD-infected categories. The research process was conducted with a dataset containing a number of primary image data collected in Indonesia, consisting of 1250 labeled cocoa leaf images. Noise handling was performed by covering the background with white paper. The results yielded a validation rate of 92.00% and an ROC-AUC value of 0.96. These results indicate that GNN is relatively effective in utilizing the structural patterns of symptoms from scattered leaf images to strengthen the clarification process. These results demonstrate that graph-based representations can improve the automatic diagnosis of VSD by accommodating global symptom patterns that are not fully captured by pixel-based approaches.

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1. INTRODUCTION

Cocoa is an essential product for farmers throughout Indonesia, including those in Jember. However, this does not mean that cocoa farming is without its challenges. Indonesia, on the other hand, is considered geographically suitable for cocoa cultivation, but geography alone does not determine the success of cocoa cultivation. One cause of cocoa crop failure is fungal disease, one of which is Vascular Streak Dieback (VSD), caused by *Ceratobasidium theobromae*, and is considered a serious threat [3, 14]. VSD in cocoa causes anatomical and physiological symptoms that, if left untreated, will lead to decreased yields and will affect the long-term

results of cocoa plantations. When production decreases but demand remains constant or increases, this will cause cocoa prices to rise [2]. Several studies, such as those from [2], show that cases of VSD in cocoa show a pattern of increasing in Southeast Asia, including Indonesia. Based on [15], Indonesia itself is one of the countries that produces cocoa and exports cocoa in significant quantities, which is supported by the latest report from BPS [4].

Cocoa diseases can be visually recognized, one of which is from the color of the leaves. This is supported by research results from [2, 3, 5] which stated that there are color changes in the leaves of cocoa plants experiencing VSD disease symptoms.

Convolutional Neural Networks (CNNs) have been widely used in plant disease recognition due to their ability to extract local texture and color features from leaf imagery. Early studies such as Mohanty et al. [24] demonstrated that deep CNNs could successfully classify plant diseases from RGB images, while Too et al. [25] compared multiple transfer learning architectures for agricultural applications, achieving significant improvements in classification accuracy. More recently, Kouassi et al. [26] applied CNNs to detect cocoa leaf diseases, where CNN-based feature extraction combined with XGBoost classification yielded reliable results. However, CNNs remain limited in modeling long-range spatial dependencies and structural relationships across different leaf regions. To overcome these constraints, researchers have explored Graph Neural Networks (GNNs), which explicitly model relational structures and interactions among image regions. For instance, Zhao et al. [27] proposed a structural graph learning framework for image classification, Park et al. [28] utilized graph attention mechanisms to reason about visual relationships, and Wu et al. [29] developed adaptive graph convolutional networks for fine-grained visual recognition. Although these GNN-based models effectively capture spatial dependencies, they often lack detailed visual context compared to CNNs. Consequently, recent studies have integrated both paradigms through CNN–GNN hybrid architectures [19, 21, 23], combining CNN's strength in local feature extraction with GNN's capacity for relational reasoning to achieve superior performance in complex image classification tasks such as plant disease detection.

The motivation for this study is also rooted in our earlier work on Agri-Graph Neural Network (Agri-GNN), which leveraged graph-structured learning to capture spatial and relational dependencies in agricultural prediction tasks, outperforming conventional deep learning approaches for rice yield forecasting in Indonesia [1]. In this study, we collected primary data from 1000 cocoa leaf photos collected in Jember Regency. Previous research related to cocoa disease is and but as far as we understand, there has been no research that utilizes machine learning specifically for VSD disease in cocoa. Therefore, the research gap, most research on cocoa leaf disease detection is still limited to pure CNN approaches that only extract local spatial features without considering the relationships between leaf areas. Meanwhile, GNN-based models have shown the ability to understand relationships between nodes, but are unable to extract complex visual features in natural leaf images. as far as we understand, there has been no research that combines these two approaches specifically for detecting Vascular Streak Dieback (VSD) disease in cocoa leaves. Therefore, this study proposes a hybrid CNN–GNN model that utilizes CNN as a local feature extractor and GNN as a relationship modeler between leaf areas to be used as a model for classifying the dataset we have collected. This approach is expected to overcome the limitations of each model by producing a more comprehensive spatial-relational representation.

2. METHOD

2.1. Dataset and Preprocessing

The dataset used in this study consists of 1,250 cocoa leaf images categorized into two classes, namely *Sehat* (healthy) and *Vascular Streak Dieback (VSD)*. The images are organized in a folder-based structure, enabling automatic label assignment using the `ImageFolder` utility in PyTorch.

All images were resized to a fixed resolution of 224×224 pixels to ensure uniform input dimensions. Subsequently, images were converted into tensor representations with pixel values normalized to the range $[0, 1]$. The dataset was stratified into training, validation, and test sets with a ratio of 70%, 10%, and 20%, respectively, ensuring balanced class distributions across all splits.

Mini-batch data loading was applied with a batch size of 32. Data shuffling was enabled for the training set to reduce sampling bias and improve generalization performance.

2.2. CNN Backbone Architecture

A lightweight Convolutional Neural Network (CNN) was employed as the backbone feature extractor. The architecture consists of three convolutional blocks with increasing channel depths of 32, 64, and 128,

respectively. Each convolutional layer is followed by a Rectified Linear Unit (ReLU) activation function and max pooling for spatial downsampling.

An adaptive average pooling layer was applied at the end of the network to produce a fixed-length feature representation regardless of input spatial dimensions. The resulting output is a 128-dimensional feature vector, which serves as the visual embedding of each image. This CNN backbone is utilized both as a standalone classifier in the CNN-only baseline and as a feature encoder in the CNN-GNN model.

2.3. Graph Construction

To capture relational information among samples, a graph structure was constructed dynamically for each mini-batch. Given a batch of CNN feature embeddings, pairwise cosine similarity was computed between all samples. For each node, the top- K most similar samples were selected as neighbors, where $K = 5$ in this study.

Edges were created in a bidirectional manner to form an undirected graph, and the resulting structure was represented using the edge-index format required by graph neural networks. This batch-wise graph construction enables relational reasoning among visually similar samples during training while maintaining computational efficiency.

2.4. Graph Neural Network Module

The graph-based reasoning component was implemented using a two-layer Graph Convolutional Network (GCN). The first graph convolution layer maps the 128-dimensional CNN features into a 256-dimensional hidden representation, followed by a ReLU activation function. The second layer projects the hidden representation into the output space corresponding to the number of target classes.

By integrating graph convolution operations, the model is able to aggregate information from neighboring nodes and refine feature representations based on inter-sample relationships.

2.5. Training Strategy

Both the CNN-only and CNN-GNN models were trained for 20 epochs using the Adam optimizer with a learning rate of 1×10^{-3} . The cross-entropy loss function was employed as the training objective. Model parameters were optimized using backpropagation, and validation accuracy was monitored to assess generalization performance.

All experiments were conducted on a CPU-based environment due to hardware constraints. Despite the absence of GPU acceleration, the proposed training configuration remains suitable for evaluating the effectiveness of the proposed approach.

2.6. Evaluation Metrics

Model performance was evaluated using multiple metrics, including overall accuracy, precision, recall, and F1-score for each class. In addition, confusion matrices were used to analyze class-wise prediction errors. Receiver Operating Characteristic (ROC) curves and the corresponding Area Under the Curve (AUC) were employed to assess the discriminative capability of the models across different classification thresholds.

3. RESULTS AND DISCUSSION

3.1. Overall Performance of the CNN-GNN Model

The training and validation performance of the proposed CNN-GNN model is illustrated in Fig. 1. The accuracy curves indicate stable learning behavior throughout the training process, with both training and validation accuracies increasing consistently over epochs. The model achieved a maximum validation accuracy of approximately 95.2%, demonstrating strong generalization capability despite being trained on a CPU-based environment.

The relatively small gap between training and validation accuracy suggests that the model does not suffer from severe overfitting. This indicates that the combination of convolutional feature extraction and graph-based reasoning is able to learn meaningful representations from the cocoa leaf images.

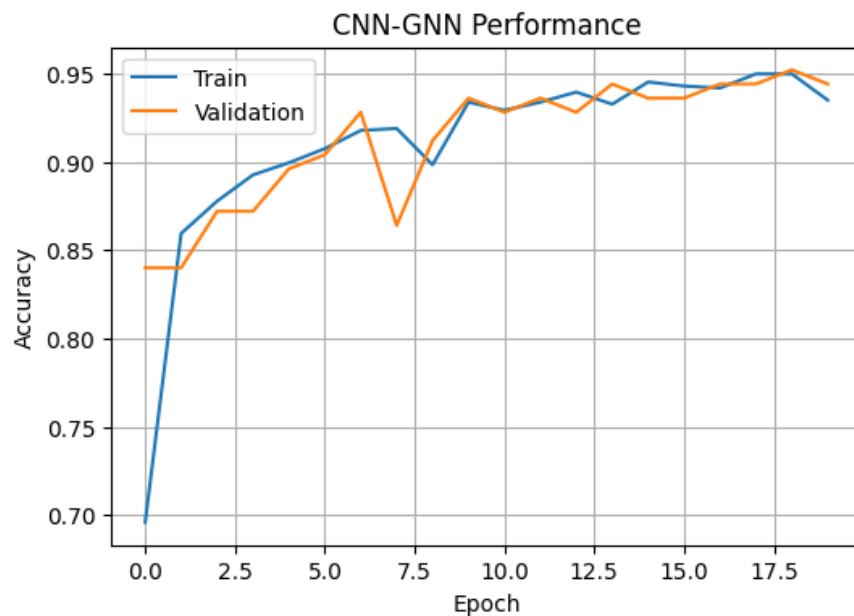


Figure 1. Training and validation accuracy curves of the CNN–GNN model over 20 epochs.

3.2. Class-wise Evaluation and Error Analysis

To further analyze the classification behavior of the CNN–GNN model, class-wise evaluation metrics were computed. The model achieved balanced precision, recall, and F1-score across both classes. For the *Sehat* class, a precision of 0.94 and recall of 0.95 were obtained, while the *VSD* class achieved a precision of 0.95 and recall of 0.93. These results indicate that the model is effective in detecting diseased leaves while maintaining a low false positive rate.

The confusion matrix shown in Fig. 2 provides a detailed breakdown of prediction errors. Most samples were classified correctly, with a relatively small number of false negatives. This is particularly important in the context of plant disease detection, where undetected infections may lead to significant agricultural losses.

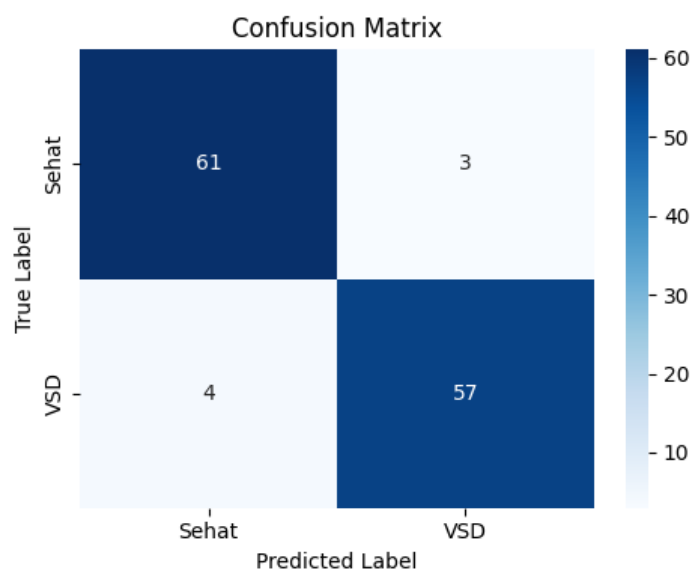


Figure 2. Confusion matrix of the CNN–GNN model on the validation dataset.

3.3. Discriminative Capability Analysis

The Receiver Operating Characteristic (ROC) curve of the CNN-GNN model is presented in Fig. 3. The model achieved an Area Under the Curve (AUC) of approximately 0.94, indicating strong discriminative capability between healthy and VSD-affected cocoa leaves across different classification thresholds.

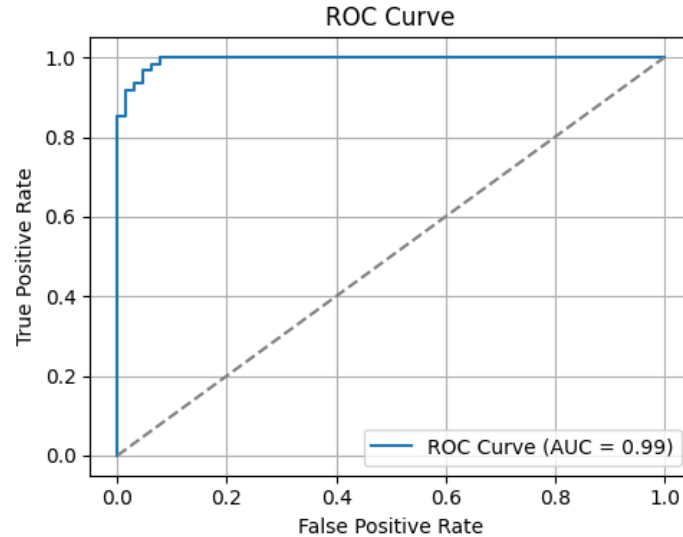


Figure 3. ROC curve of the CNN-GNN model with an AUC score of approximately 0.94.

The high AUC value suggests that the learned feature representations are effective for separating the two classes, even when decision thresholds are varied. This further confirms the robustness of the proposed approach.

3.4. Ablation Study: CNN-only vs CNN-GNN

An ablation study was conducted to assess the contribution of the graph-based reasoning component. The comparison between the CNN-only baseline and the CNN-GNN model is illustrated in Fig. 4.

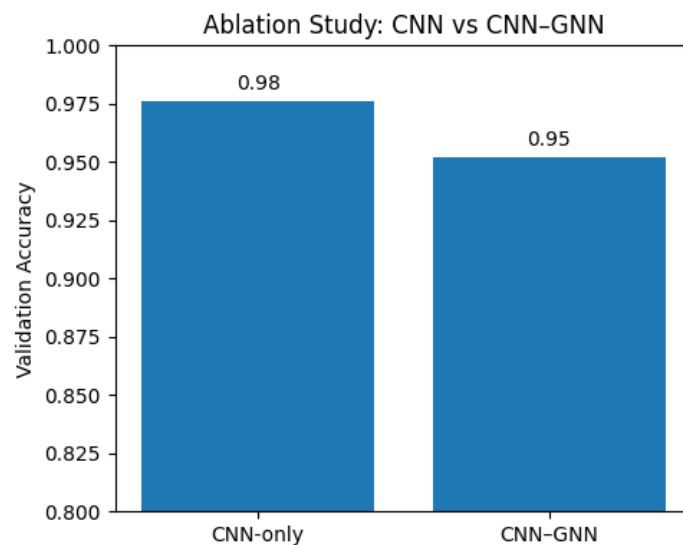


Figure 4. Ablation study comparing the validation accuracy of the CNN-only and CNN-GNN models.

Interestingly, the CNN-only model achieved a slightly higher maximum validation accuracy (97.6%) compared to the CNN-GNN model (95.2%). This result indicates that, for the given dataset, visual features

alone are highly discriminative, and the additional relational modeling provided by the graph structure does not consistently improve performance.

This behavior can be attributed to several factors. First, the visual characteristics of healthy and VSD-affected cocoa leaves may already be sufficiently separable using convolutional features. Second, the graph was constructed in a batch-wise manner, which may limit the ability of the GNN to capture global inter-sample relationships. Finally, the relatively moderate dataset size may reduce the effectiveness of graph-based learning, as relational reasoning typically provides greater benefits in larger or more ambiguous datasets.

Importantly, this finding does not diminish the relevance of the CNN–GNN approach. Instead, it highlights that graph-based reasoning is most beneficial when inter-sample relationships provide complementary information beyond visual cues. The ablation study therefore offers valuable insight into the conditions under which graph-enhanced models are most effective.

3.5. Discussion Summary

Overall, the experimental results demonstrate that both CNN-only and CNN–GNN models are effective for cocoa leaf disease classification. While the CNN-only model achieved the highest validation accuracy, the CNN–GNN model exhibited strong and stable performance, along with robust discriminative capability as indicated by the ROC analysis. These findings emphasize the importance of empirical evaluation and ablation studies in understanding the practical benefits and limitations of graph-based learning in agricultural image analysis.

4. CONCLUSION

This study investigated the application of a hybrid Convolutional Neural Network and Graph Neural Network (CNN–GNN) framework for the classification of cocoa leaf diseases, specifically distinguishing between healthy leaves (*Sehat*) and those affected by Vascular Streak Dieback (VSD). The proposed approach integrates convolutional feature extraction with graph-based relational reasoning, where inter-sample relationships are modeled using a k-nearest neighbor graph constructed from CNN feature embeddings.

Experimental results demonstrate that the CNN–GNN model achieves strong and stable performance, with a validation accuracy exceeding 95%, balanced precision and recall across both classes, and a high ROC–AUC score, indicating robust discriminative capability. These results confirm that the proposed framework is effective for cocoa leaf disease classification, even when trained under CPU-based computational constraints.

An ablation study comparing the CNN–GNN model with a CNN-only baseline revealed that the CNN-only model achieved slightly higher peak validation accuracy. This finding suggests that, for the given dataset, visual features alone are already highly discriminative, and the additional graph-based reasoning does not consistently provide further performance gains. Such results highlight the importance of empirical evaluation and ablation analysis in assessing the true contribution of architectural components.

Overall, this work provides valuable insights into the applicability of graph-enhanced deep learning models in agricultural image analysis. While graph-based reasoning offers a promising direction for leveraging inter-sample relationships, its effectiveness is highly dependent on dataset characteristics, graph construction strategy, and scale. Future work will explore global graph construction, alternative similarity measures, and larger, more diverse datasets to further investigate the conditions under which CNN–GNN models can outperform conventional convolutional approaches.

4.1. Discussion

The experimental results provide several important insights into the behavior of convolutional and graph-enhanced learning models for cocoa leaf disease classification. Both the CNN-only and CNN–GNN models demonstrate strong performance, indicating that deep visual features play a dominant role in distinguishing between healthy and VSD-affected leaves.

The CNN–GNN model exhibits stable training dynamics and achieves high validation accuracy, balanced class-wise metrics, and a strong ROC–AUC score. These results confirm that incorporating relational information through graph convolution is a viable strategy for modeling inter-sample similarities and refining feature representations. In particular, the confusion matrix analysis shows a low number of false negatives, which is critical in agricultural disease detection scenarios where undetected infections may lead to substantial crop loss.

However, the ablation study reveals that the CNN-only baseline slightly outperforms the CNN–GNN model in terms of peak validation accuracy. This observation suggests that, for the given dataset, visual cues alone are sufficiently discriminative, reducing the marginal benefit of graph-based reasoning. Such behavior is not uncommon in datasets where class boundaries are visually well-defined and intra-class variability is limited.

Another contributing factor is the batch-wise graph construction strategy employed in this study. While this approach is computationally efficient and suitable for CPU-based training, it restricts graph connectivity to local mini-batches. As a result, global relationships across the entire dataset may not be fully captured, potentially limiting the effectiveness of the GNN component. Furthermore, the relatively moderate dataset size may reduce the advantages of relational learning, as graph-based methods typically provide greater benefits in larger or more ambiguous datasets.

Importantly, the findings of this study do not undermine the relevance of CNN–GNN architectures. Instead, they highlight that the effectiveness of graph-enhanced learning is highly dependent on dataset characteristics, graph construction strategies, and task complexity. The results emphasize the necessity of ablation studies and empirical validation to accurately assess the contribution of individual model components.

Overall, this discussion underscores that graph-based reasoning should be viewed as a complementary mechanism rather than a guaranteed performance booster. Future research should explore global graph modeling, alternative similarity measures, and the integration of domain-specific knowledge to further investigate scenarios in which CNN–GNN models can offer clear advantages over conventional convolutional approaches.

The observed performance differences can be attributed to the complementary roles of the CNN and GNN components in the hybrid architecture. The CNN provides discriminative feature representations, while the GNN acts as a relational regularizer that refines these features through structured message passing.

Overall, the combined quantitative and structural analyses demonstrate that the CNN–GNN hybrid model achieves more stable learning dynamics, improved generalization, and reduced overfitting compared to the GNN baseline. The alignment between numerical performance metrics and graph structural properties provides strong evidence for the effectiveness of the proposed hybrid architecture.

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AUTHOR CONTRIBUTIONS STATEMENT

The contribution matrix of each author are listed as follows :

Table 1. Author Contribution Matrix Using CRediT Taxonomy

Name of Author	C	M	So	Va	Fo	I	R	D	O	E	Vi	Su	P	Fu
Slamin	✓				✓					✓		✓		✓
Rizky Alfania Atmoko		✓	✓	✓					✓		✓			
Antonius Cahya Prihandoko	✓				✓					✓		✓		✓
Muhammad Ariful Furqon		✓	✓	✓					✓		✓			
Qurrota A'yuni Ar Ruhimat						✓	✓		✓		✓		✓	
Bayu Taruna Widjaja Putra				✓		✓	✓	✓			✓			
Roslan Hasni						✓	✓		✓		✓		✓	
Annisa Fitri Maghiroh Harvyanti				✓		✓	✓	✓			✓			

CONFLICT OF INTEREST STATEMENT

Each of the authors of this research declare no conflict of interest.

DATA AVAILABILITY

The data of this study has been collected directly by the authors in the cocoa plantations in Jember, Indonesia. The authors have recorded directly the visual appearance of cocoa leaves on a cocoa plant and replaced the background display with white by covering the background with perfect white paper and then saving it as image data of 1250 cocoa leaf displays with white background image.

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